

TASK 1 REPORT

ZOMATO RESTAURANT DATA ANALYSIS

Data Science Internship

Submitted By:

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Domain: Data Science

Task Number: 1

Tools & Technologies Used

Python | Pandas | NumPy | Matplotlib | Seaborn | WordCloud | Google
Colab

EXECUTIVE SUMMARY

This project analyzes the Zomato restaurant dataset to understand customer preferences, restaurant ratings, cuisine popularity, location-wise restaurant distribution, and pricing trends. The dataset was cleaned by handling missing values, removing duplicate and corrupted records, and converting important columns into appropriate data types. Exploratory Data Analysis (EDA) was performed using Python libraries such as Pandas, Matplotlib, Seaborn, and WordCloud. Several visualizations were created to identify patterns and relationships within the data. Based on the findings, business recommendations were proposed for a restaurant discovery platform.

1. OBJECTIVE

The objective of this project is to analyze restaurant and review data to extract meaningful insights related to restaurant ratings, cuisine preferences, location hotspots, pricing patterns, and customer behavior. The analysis aims to support data-driven business decisions and improve customer recommendations.

2. DATASET DESCRIPTION

Dataset Name: Zomato Restaurant Dataset

Source: Kaggle

Total Records: 56,252

Total Features: 13

Main Attributes

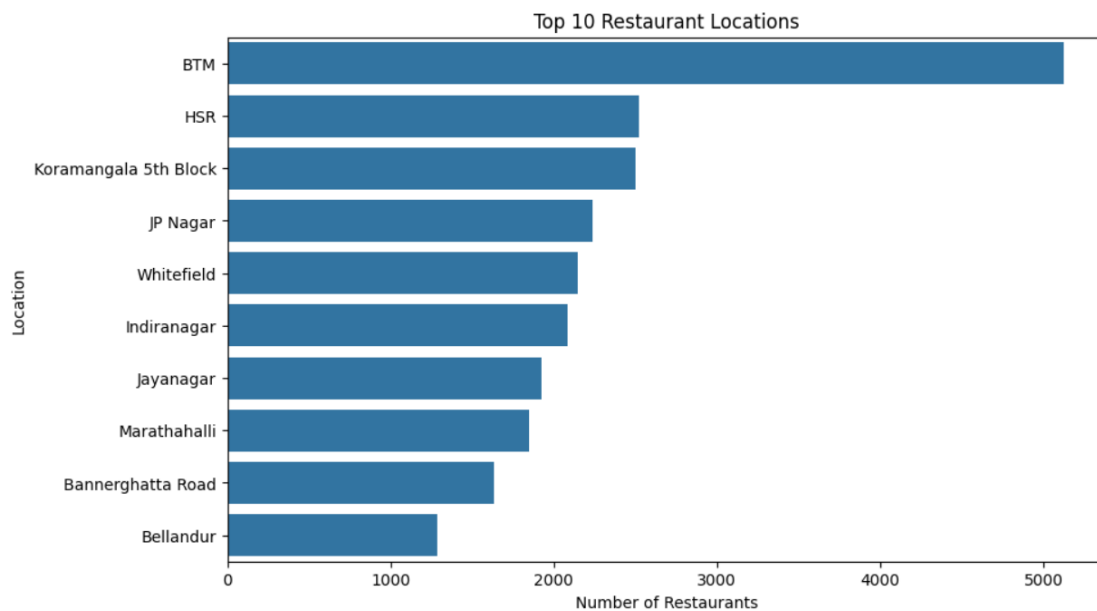
- Restaurant Name
- Address
- Location
- Online Order
- Book Table
- Rating
- Votes

- Restaurant Type
- Cuisines
- Approximate Cost for Two People
- Dish Liked
- Listed Category

3. DATA CLEANING

- Loaded the dataset into Google Colab.
- Examined the dataset structure and data types.
- Identified missing values.
- Removed duplicate records.
- Removed corrupted rows with misplaced text.
- Converted the **Rate** column into numeric format.
- Converted the **Approximate Cost for Two People** column into numeric format.
- Cleaned categorical columns used for analysis.

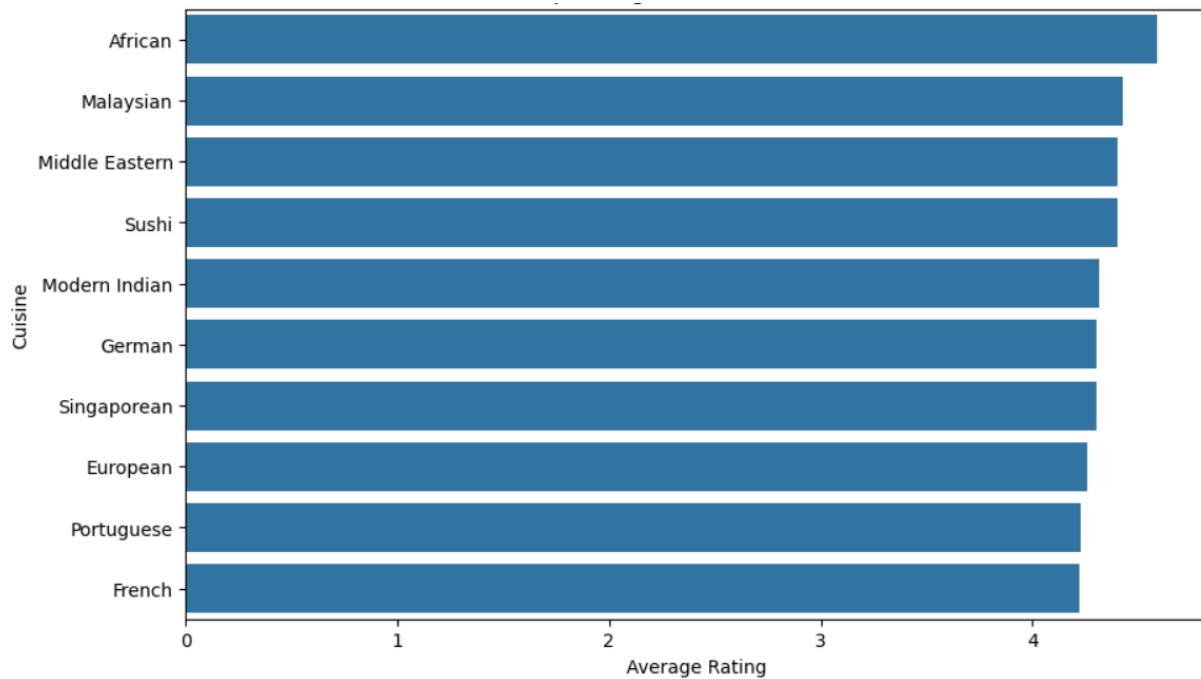
4. EXPLORATORY DATA ANALYSIS



Visualization 1 – Top Restaurant Locations

Observation

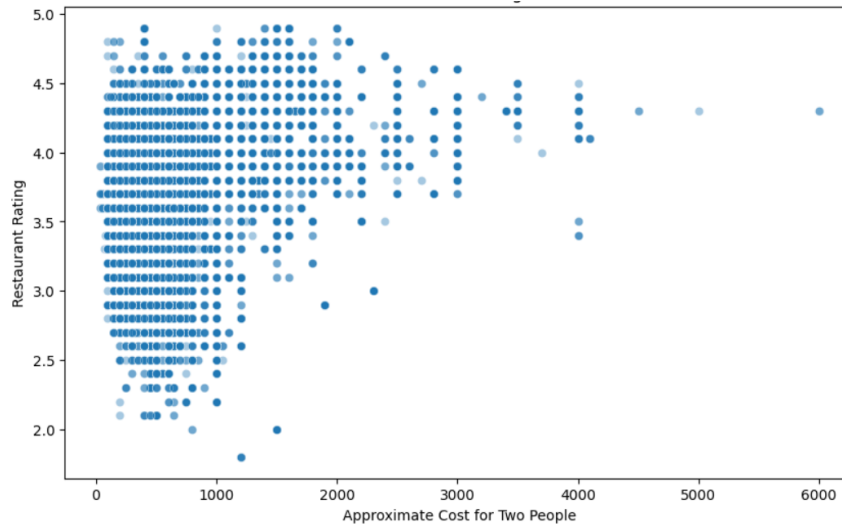
BTM has the highest concentration of restaurants, followed by HSR and Koramangala. These locations represent major restaurant hubs with high customer activity.



Visualization 2 – Top Rated Cuisines

Observation

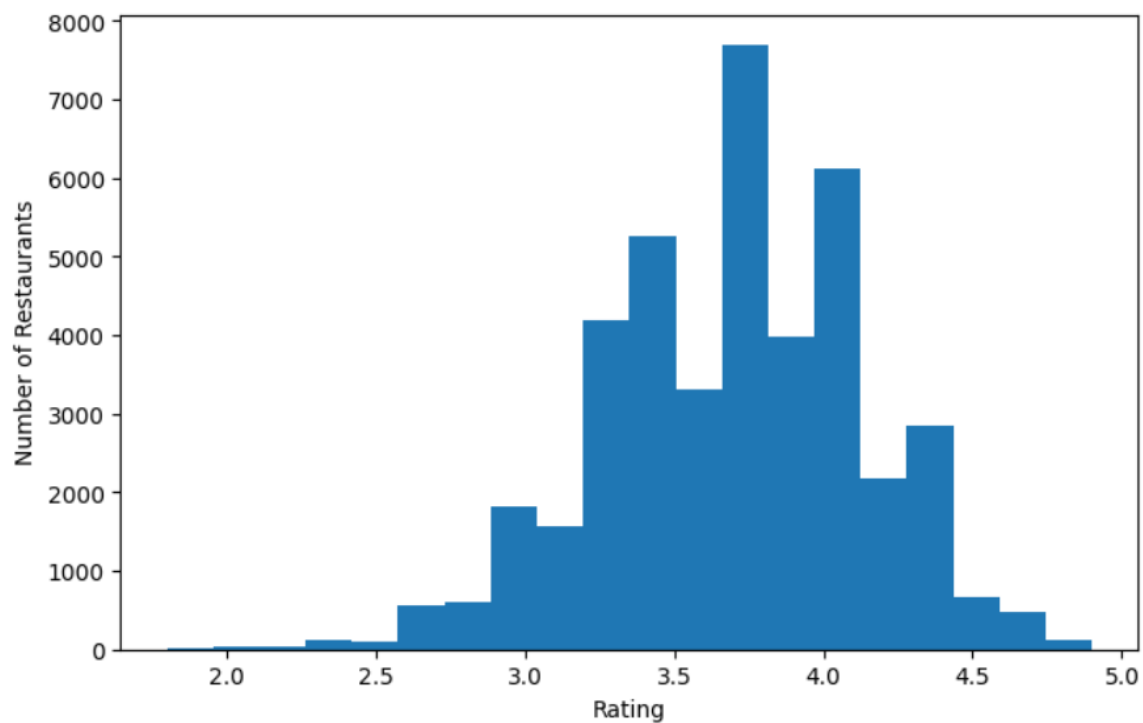
African, Malaysian, and Middle Eastern cuisines have the highest average ratings in the dataset. These cuisines receive consistently positive customer feedback.



Visualization 3 – Price vs Rating

Observation

Most restaurants charge between ₹200 and ₹800 for two people. The analysis shows that higher-priced restaurants do not necessarily receive higher ratings.



Visualization 4 – Rating Distribution

Observation

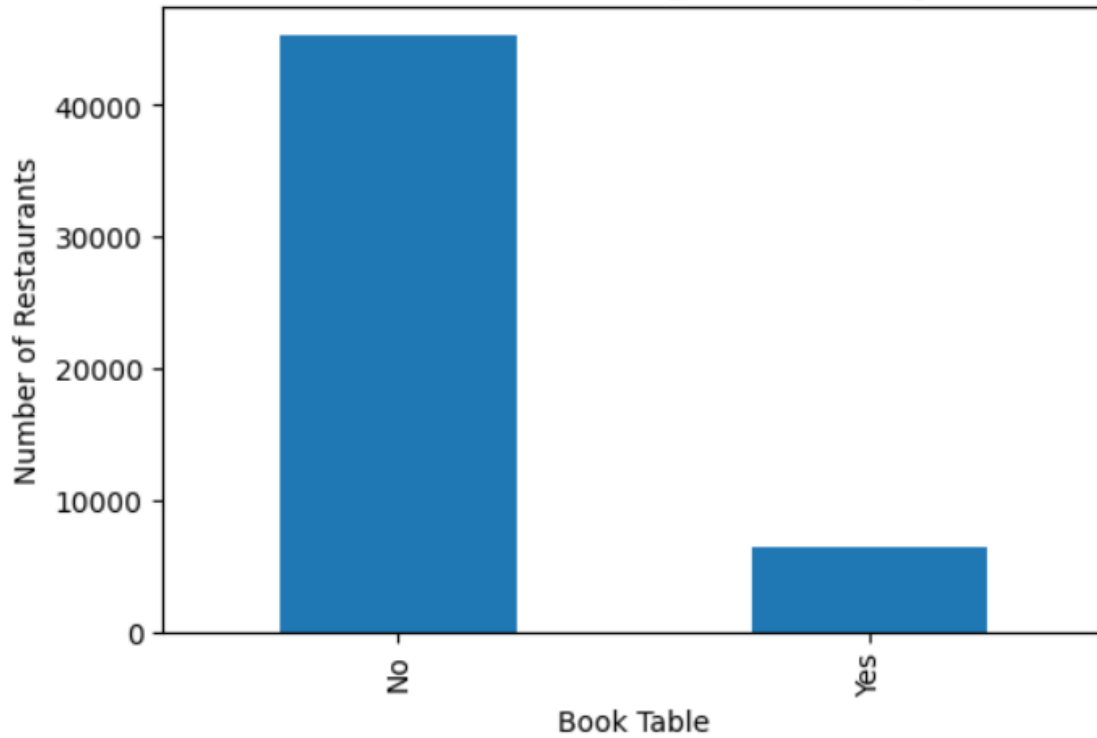
The majority of restaurants have ratings between 3.5 and 4.2, indicating generally positive customer satisfaction.



Visualization 5 – Online Order Analysis

Observation

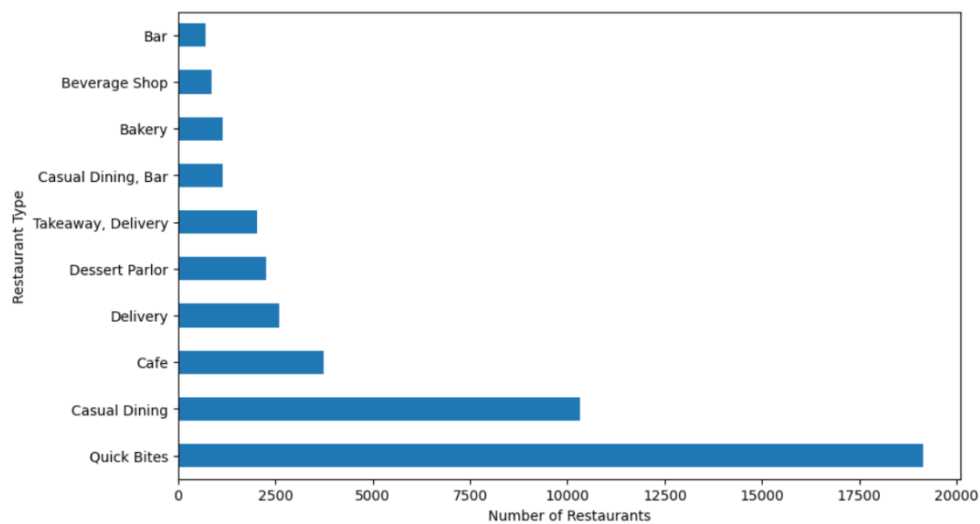
Most restaurants provide online ordering services, reflecting the growing importance of food delivery platforms.



Visualization 6 – Table Booking Analysis

Observation

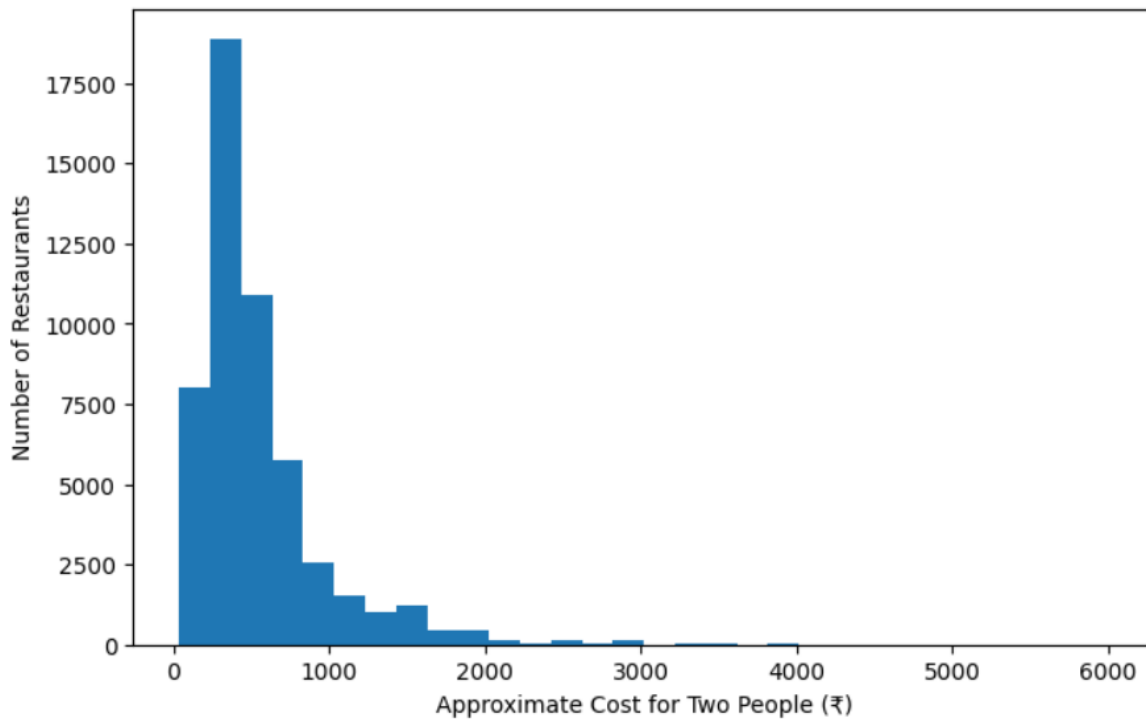
Only a smaller percentage of restaurants offer table booking facilities. This feature is more common among premium dining establishments.



Visualization 7 – Restaurant Type Distribution

Observation

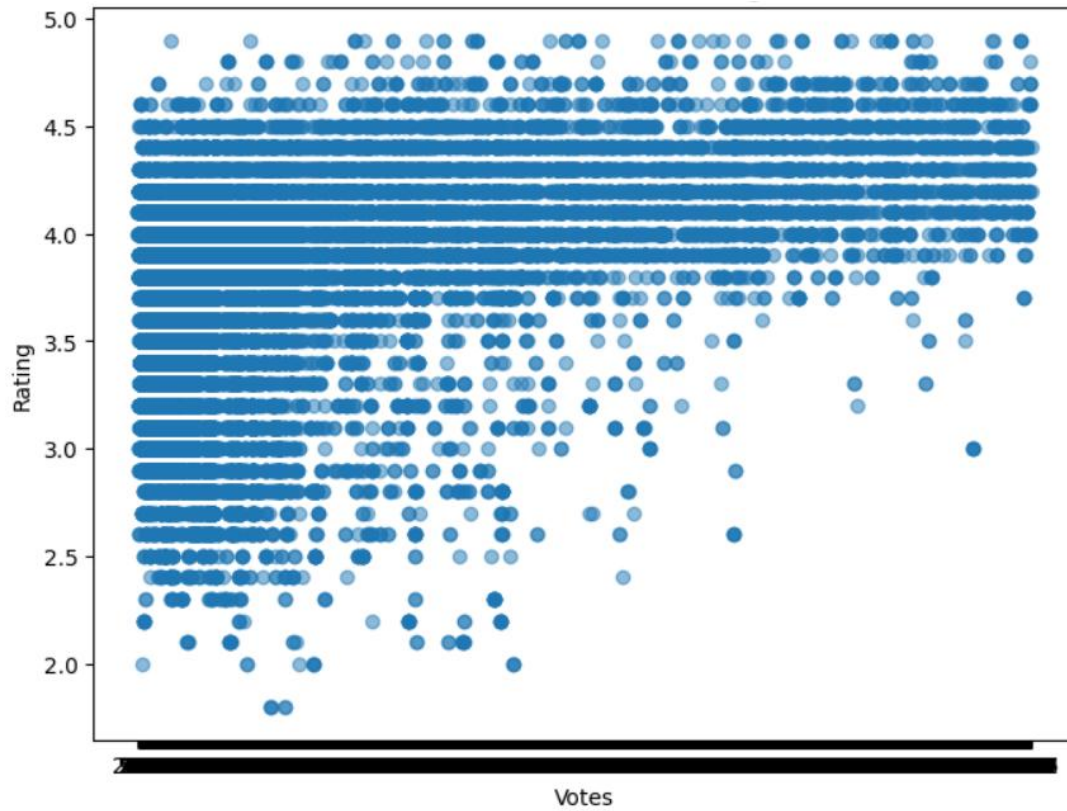
Quick Bites and Casual Dining are the most common restaurant types in the dataset, followed by Cafés.



Visualization 8 – Cost Distribution

Observation

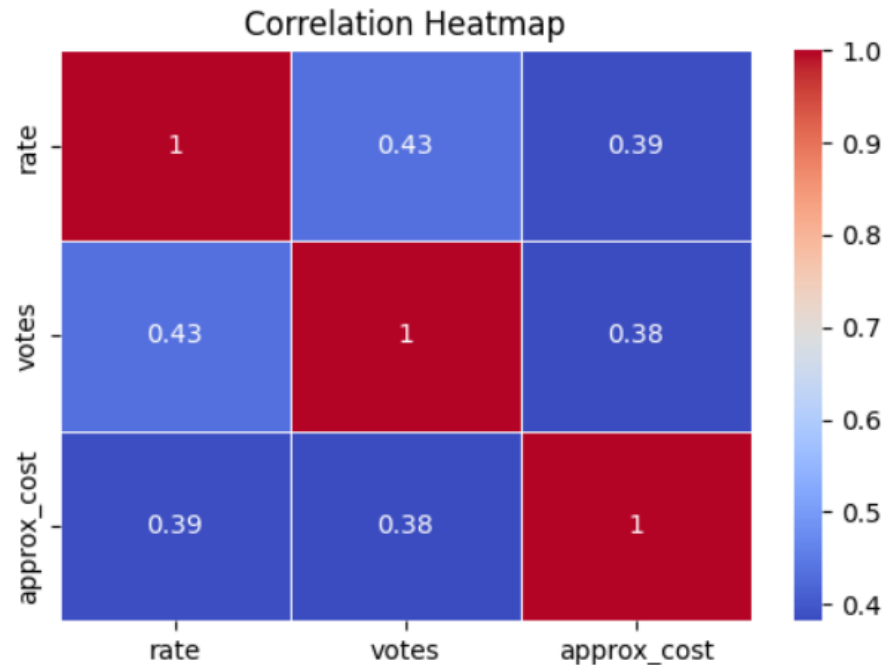
The majority of restaurants fall into the affordable pricing range, making them accessible to a wide customer base.



Visualization 9 – Votes vs Rating

Observation

Restaurants with higher ratings generally receive more customer votes, indicating a positive relationship between popularity and customer satisfaction.



Visualization 10 – Correlation Heatmap

Observation

The heatmap shows a weak positive relationship between restaurant ratings and votes, while the relationship between ratings and cost is relatively small.



Visualization 11 – Popular Cuisines WordCloud

North Indian, Chinese, South Indian, Fast Food, and Biryani appear as the most frequently offered cuisines.



Observation

Popular dishes such as pizza, burgers, biryani, coffee, sandwiches, and pasta appear frequently, highlighting customer preferences.

- BTM has the highest number of restaurants.
- North Indian and Chinese cuisines are among the most popular.
- Most restaurants provide online ordering.
- Table booking is available only in a limited number of restaurants.
- Restaurant ratings are generally high, with most ratings between 3.5 and 4.2.
- Restaurant price has only a weak relationship with customer ratings.
- Restaurants receiving more customer votes generally have higher ratings.

6. RECOMMENDATIONS

1. Partner with highly rated restaurants to improve platform quality.
2. Focus promotional campaigns on restaurant hotspot locations such as BTM, HSR, and Koramangala.
3. Promote popular cuisines through personalized recommendations.
4. Encourage online ordering by offering discounts and promotional campaigns.
5. Use customer ratings and preferences to develop personalized restaurant recommendations.

7. CONCLUSION

This project successfully analyzed the Zomato restaurant dataset to understand customer preferences, cuisine popularity, pricing trends, and restaurant distribution. After cleaning the dataset and performing exploratory data analysis, several useful business insights were identified. The findings indicate that restaurant quality and customer satisfaction depend more on service and food quality than on price. The analysis also highlights major restaurant hotspots and popular cuisines, providing valuable recommendations for improving customer engagement and business growth.

8. PROJECT LINKS

Google Colab Notebook:

[Google Colab Notebook](#)

GitHub Repository:

[GitHub Repository](#)